



Guglielmo Grillo

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Villanuova 42, 38065, Mori (TN), Italy (Home)

● ABOUT MYSELF

I am a PhD student in Physics at the University of Trento (Statistical and Biological Physics group) with a strong passion for computer science. This passion led me to choose a computational path both for my curricular and extracurricular classes during my bachelor's and Master's degrees and to work on personal projects in my free time. I'm also one of the founding members of Progetto Apollo, a scientific divulgation group operating at the University of Trento in collaboration with Arditodesio. Lately, I've been exploring management and business. Since 2019, I've been a tutor for the University of Trento. My hobbies include action Metroidvania video games, tabletop role-playing games, and HEMA fencing.

● DRIVING LICENCE

Driving licence: B | 09/03/2017 - 30/04/2027

● WORK EXPERIENCE

20/09/2023 - 06/12/2023 - TRENTO, ITALY

SOLVER - INDUSTRIAL AI CHALLENGE ADIGE BLM GROUP / HUB INNOVAZIONE TRENTINO (HIT)

During the 11 weeks, our team collaborated with BLM Group company, the parent firm of Adige SpA, to develop an AI planner. The model can analyze sales trends and will serve as a central point for the manufacturing workflow decision-making. The organization committee decided to award us first place in the competition after evaluating the number of challenges we encountered and the ways we solved or worked around them.

Other than being responsible for data preprocessing, my role was also that of team supervisor in charge of keeping track of tasks performed by the various team members and communications with the company and the organization committee. This allowed me to gain a better understanding of how a data science project is carried out in a real-world scenario, to appreciate the importance of time and task management, and the necessity of team investment to reach a common goal.

Business or sector Professional, scientific and technical activities | **Website:**

<https://www.trentinoinnovation.eu/en/innovate/innovation-tools/ai-challenge/>

2017 - 10/2022 - TRENTO, ITALY

STORYTELLER ARDITODESIO / UNIVERSITY OF TRENTO

I'm one of the original members of Progetto Apollo, a scientific divulgation group operating at the University of Trento in collaboration with Arditodesio. Some of our activities include research on a key topic or an interview with a Professor at the University of Trento to write narratives with the principles of storytelling. The narratives were then performed in various editions of Teatro della Meraviglia, EIT RAW Materials 2021, CO.SCIENZA Festival 2019, Focus Days 2019 and Fisicità 2017. In 2022 we partnered with Caltech for the event "La Terra fatta dagli umani" (Earth made by humans) about the beginning of the Anthropocene. The narratives were presented both in Trento and at Caltech.

06/07/2015 - 31/07/2015 - TRENTO, ITALY

INTERNSHIP FBK - FONDAZIONE BRUNO KESSLER

The task of our two-student team was to study the effects of different NaCl concentrations on the contact angle between water droplets and various substrates (Borosilicate, Aluminium, and Soda-Lime Glass). The analysis was conducted both in the static regime, where droplets were studied at deposition and in the dynamic regime where we analyzed the contact angle as a function of time. The experience ended with a presentation held in the context of "Notte dei Ricercatori 2015".

09/2019 - CURRENT - TRENTO, ITALY

TUTOR UNIVERSITY OF TRENTO

In the in-class courses, my task as a tutor was to prepare ad-hoc exercises beforehand and then follow the students in their reasoning by providing hints and guidance toward the solution. In the laboratory courses, our job was to explain the lab experiences to the students and assist them in the data collection and analysis ensuring that they understood the underlying reasoning and scientific approach.

I was assigned to the following courses:

- Feb 2025 - June 2025: Scientific Programming (teacher assistant, DF)
- Feb 2024 - June 2024: Scientific Programming (teacher assistant, DF)
- Sep 2022 - Dec 2022: Introductory Physics Laboratory (DII), Mathematical Basis for Cognitive Sciences (CIMEQ), Mathematics and Statistics I (CIBIO);
- Mar 2022 - June 2022: Laboratory of Physics I with Matlab (DF), Introductory Physics Laboratory (DICAM);
- Sep 2021 - Dec 2021: Introductory Physics Laboratory (DII);
- Sep 2020 - Dec 2020: Mathematics and Statistics (A3C);
- Sep 2019 - Jan 2020: Calculus I (DICAM), Calculus I (DIPSCO), Quantitative Methods for life-sciences (DIPSCO);

The underlined courses represent the courses for which I was the only tutor. In all other courses, I was part of a team of 2 to 4 tutors.

16/09/2024 - 04/12/2024 - TRENTO, ITALY

WORKSHOP ORGANIZER PHYSICS DEPARTMENT - UNIVERSITY OF TRENTO

I and three other PhD students in the department organized the 2024/2025 edition of the Physics PhD Workshop, an annual event where PhD student can share their ongoing work and look for collaborations inside the department. Our main tasks were:

- Choose and contact a Keynote speaker ("Lectio Magistralis") for the event;
- Keep the department informed of organization, schedule, and deadlines;
- Organize the coffee breaks, the social dinner, and the poster session;
- Develop a website for the event.

Website: <https://ppw2024.physics.unitn.it/>

03/05/2025 - 04/05/2025 - MORI (TN), ITALY

POLLING STATION SUPERVISOR (SEGRETARIO DEL SEGGIO ELETTORALE)

I was elected secretary for the local elections in my hometown. My job was to keep track of all events and procedures occurring during the election day, record the voting process and the results of the elections in pooling station I was assigned to. I worked in close contact with the president to ensure the correct proceedings of the voting and final voting count and with the scrutineer to validate and record any non standard request from the voters.

30/06/2025 - 07/07/2025 - TRENTO, ITALY

YOUNG RESEARCHER INTERN - IPSP OPTOI / DEPARTMENT OF PHYSICS

During Industrial Problem Solving with Physics (IPSP), our team of 7 PhD and Master students from physics and computer science solved the challenge proposed by the company Optoi. In this timespan, *we were able to speed up the production of a company asset by 44% on average and by 53% in the most common scenario*. We achieved this improvement by combining a theoretical understanding of the problem that we implemented via a

computer program and practical insight obtained via experiments performed in the company.

Website: <https://www.optoi.com/it>

● CONFERENCES & SEMINARS

04/05/2023 - 05/05/2023 University of Trento

Statistical Physics of Living Systems workshop This two-day workshop gathered the members of the SBP group at the University of Trento and the QLS group of ICTP in Trieste to present and discuss the work carried out within both groups. The workshop aimed to foster tighter collaborations, address open problems, and develop strategies for future joint activities. My contribution consisted of presenting my master's thesis work.

Links <https://sbp.physics.unitn.it/statistical-physics-of-living-systems-workshop>

04/12/2024 Trento

Physics PhD Workshop 2024/2025 The Physics PhD Workshop (PPW) is an annual scientific event at the University of Trento. During this event, second year PhD students from various fields can come together to share their work and discuss the latest developments in their respective areas of research.

My contribution consisted in the organization of the event and a presentation of my PhD work "Coarse Grained Simulations of Kinetoplast DNA".

Links <https://ppw2024.physics.unitn.it/>

● EDUCATION & TRAINING

26/01/2026 - 30/01/2026

FROM RESEARCH TO BUSINESS: A KNOWLEDGE TRANSFER APPROACH- FONDAZIONE HIT IN COLLABORAZIONE CON UNIVERSITÀ DI TRENTO

During the week, we explored the connection between our research and business, focusing on stakeholders, value proposition, business models, and problem-solution-market fit. The course comprises both theoretical lectures and hands-on classes, in which we applied the concepts learned to our research project. Instead of a standard exam, we opted for a 30min discussion on the key insight gained during the course and possible follow-ups.

Field(s) of study: Inter-disciplinary programmes and qualifications involving business, administration and law | **Website:** <https://www.trentinoinnovation.eu/partecipa/formazione-per-il-dottorato/>

01/11/2023 - Current - ITALY

PHD IN PHYSICS- UNIVERSITY OF TRENTO

My PhD project revolves around two major parallel topics: chromatin organization and Kinetoplast DNA.

The main goal of the project is the design of a polymer model to describe chromatin re-organization: the simulated data, in conjunction with experimental ones, will be used in a machine learning pipeline to predict breast cancer progression.

Simultaneously, I am also developing a coarse-grained model of *Crithidia fasciculata* DNA focused on its topological properties.

As a side project, I am working on a technical article about best practices to organize a computational project.

During my PhD I followed the courses:

- Scientific Writing, Speaking and Storytelling
- Multiscale modeling: from the atom to the cell
- Advance Statistical Mechanics
- Start-Up Lab
- GPU Computing

Field(s) of study: Computational Biophysics | **Level in EQF:** 8 | **Thesis:** Combining physics, biology and machine learning to understand genome organization in cancer cells (supervisor: prof. Luca Tubiana) | **Website:**

<https://sbp.physics.unitn.it/the-group/>

09/2020 - 03/2023 - TRENTO, ITALY

MASTER'S DEGREE IN PHYSICS- UNIVERSITÀ DEGLI STUDI DI TRENTO

Statistical Biophysics path with a focus on computational methods.

Relevant courses:

- Quantum Computing and Quantum Machine Learning
- Advanced Computational Physics
- Experimental Methods
- Laboratory of Advanced Electronics (signal processing via FPGA)
- Multi-Scale Methods in Soft Matter Physics
- Statistical Mechanics and Statistical Field Theory.

Field(s) of study: Physics | **Final grade:** 110/110 | **Level in EQF:** 7 | **National classification:** Laurea Magistrale | **Type of credits:** CFU | **Number of credits:** 120 | **Thesis:** Computational Models Of Astrocyte-Neuron Dynamics To Explain The Formation Of Working Memory (supervisor: prof. Luca Tubiana) | **Address:** Via Calepina, 1438122 | **Website:** <https://www.unitn.it/>

02/2021 - 03/06/2021 - ITALY

CYBERCHALLENGE.IT - CERTIFICATE- UNIVERSITÀ DEGLI STUDI DI TRENTO IN PARTNERSHIP WITH LABORATORIO NAZIONALE DI CYBERSECURITY (CINI)

During my master's degree, I took part in the CyberChallenge.IT (February-June 2021) a course designed to provide the methodological and practical basis to analyze vulnerabilities and possible attacks and identify the most suitable solutions to prevent them. Admission to the course followed the passing of three levels of tests that recorded an admission rate of 13.70%. The total duration of the course was 72 hours (24 hours of lecture and 48 hours of hands-on experiences) followed by a local final competition lasting 7 hours.

Field(s) of study: Information and Communication Technologies | **Website:** <https://cyberchallenge.it/halloffame/2019>

09/2016 - 03/2020 - TRENTO, ITALY

BACHELOR'S DEGREE- UNIVERSITÀ DEGLI STUDI DI TRENTO

Relevant courses:

- Scientific Computing (fka Computer Science), Introduction to Machine Learning (fka Advanced Algorithms)
- Linear Algebra I, Analysis I, II, III, and Complex Analysis, Complementary Mathematics for Quantum Mechanics
- Physics I, II, and III, Analytical Mechanics, Quantum Mechanics
- Physics Laboratory I (measures), II (analogue electronics and optics), and III (digital electronics)
- Chemistry with Laboratory Sessions.

I took part in a Scientific Storytelling course twice during my bachelor's degree. This resulted in two 5-minute plays and the creation of the storytelling group Progetto Apollo. After graduating, I took part in the course Data Science for Physicists as a listener (March-June 2020).

Field(s) of study: Physics | **Level in EQF:** 6 | **National classification:** Laurea Triennale | **Type of credits:** CFU | **Number of credits:** 180 | **Thesis:** Emergent features: from renormalization group to artificial intelligence (supervisor: prof. Raffaello Potestio) | **Address:** Via Calepina, 1438122 | **Website:** <https://www.unitn.it/>

2011 - 2016 - ROVERETO, ITALY

SCIENTIFIC HIGH SCHOOL DIPLOMA- LICEO ANTONIO ROSMINI (ROVERETO, TN)

Chemistry, Physics, Latin (Language and Literature), Mathematics, Biology, History, Ethics, English (Language and Literature), and Italian (Language and Literature).

Personal accomplishment:

- Qualification to the regional phase of Chemistry Olympics and Physics Olympics.
- I took part in "Insieme per studiare" (tutoring at the high school level) for two years as a tutor.
- Two seven-day language study trips abroad (Liverpool and Broadstairs)

Level in EQF: 4 | **National classification:** Diploma Liceo Scientifico | **Thesis:** La concezione della scienza di Richard Feynman | **Address:** Corso Bettini, 86 - ROVERETO38068

02/2019 - 04/2019

INNOVATION CAMP DIPLOMA- UNIVERSITY OF TRENTO IN PARTNERSHIP WITH SAMSUNG

The first month was devoted to a 25h online course about the main topics of Industrial Innovation and a two-day intensive training course I was allowed to attend upon ranking among the best in the online test. The following months were used to develop a project work on a real business case.

Field(s) of study: Business, administration and law | **Website:**

<https://webmagazine.unitn.it/evento/ateneo/69296/samsung-innovation-camp>

20/02/2025 - 06/2025 - TRENTO, ITALY

START-UP LAB CERTIFICATE- SCHOOL OF INNOVATION - TRENTO

During the course, we learnt the basics of product development following standard practices such as the lean startup. My group started from an idea of the sector we wanted to work on, conducted some preliminary interviews of businesses, and iterated to find a useful and requested problem/solution pair. Then, we developed a mechanical prototype that tries to address the chosen problem and developed a business plan to estimate the financial viability of our idea. At the end of the initiative, we produced a physical MVP of our project and pitched the idea to the audience on the final day.

As a group leader, my task was to ensure the constant commitment of the team members and respect for the deadlines.

Field(s) of study: Business and administration not further defined , Marketing and advertising | **Thesis:** TCGs Scanner - A fast and safe way to highlight valuable cards | **Website:** <https://clabtrrento.it/it/attivita/start-up-lab>

GETTING STARTED WITH ACCELERATED COMPUTING IN CUDA C/C++-

NVIDIA [HTTPS://WWW.NVIDIA.COM/EN-US/TRAINING/](https://www.nvidia.com/en-us/training/)

● **TECHNICAL SKILLS**

Advanced User

I've been using Python 3 since 2018 for various programming purposes. My main focus is on scientific packages (NumPy and SciPy), Data Science/Machine Learning packages (Pandas, Pytorch, Sklearn), and Data Visualization (Matplotlib both for plotting and animation, Jupyter Notebook with widgets, Quarto), but I also worked on projects requiring non-scientific-oriented packages like image recognition (OpenCV's wrapper opencv-python) and text recognition packages (Tesseract's wrapper py-tesseract).

When possible, I try to implement algorithms from scratch; therefore, I'm also capable of avoiding the most frequent Python bottlenecks. If needed, I can also work with object-oriented Python and type hints.

Lately, I've been focusing on improving my C-related skills matured during the courses *Scientific Computing*, *Computational Physics*, *Advanced Computational Physics*, and the *Cyberchallenge.IT*. I'm mainly studying efficient numerical calculation, memory management, and parallelism with a focus on scientific simulations. I'm complementing these skills with readings about NVIDIA GPU architecture, CUDA, and the CUDA-C extension.

Intermediate User

Intermediate user refers to programming languages and technologies that I can understand and use but are not my main focus.

- **Programming languages:** Cuda, C++, Matlab, Java, Verilog.
- **Molecular Dynamic Engine:** LAMMPS.
- **Technologies:** Regex, SQL, Git, Linux command line (bash).
- **Markup languages:** LaTeX, Markdown (GitHub syntax).
- **Hardware:** Raspberry Pi, Arduino.

Beginner Users

Beginner user refers to programming languages and technologies that I've been exposed to and I'm confident I can learn in a short amount of time.

- **Programming languages:** Julia, Javascript, Mathematica, C# .
- **Markup languages:** Html, CSS.
- **Miscellanea:** Unity game engine, Gromacs.

LANGUAGE SKILLS

Mother tongue(s): ITALIAN

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	B2	B2

DIGITAL SKILLS TEST RESULTS

	Information and data literacy	ADVANCED	Level 6/6
	Communication and collaboration	ADVANCED	Level 5/6
	Digital content creation	ADVANCED	Level 6/6
	Safety	ADVANCED	Level 6/6
	Problem solving	ADVANCED	Level 6/6

Results from a [Self-assessment](#) Based on [The Digital Competence Framework 2.1](#).

PUBLICATIONS

[Organisation and dynamics of individual DNA segments in topologically complex genomes](#) 2025

This article has been accepted for publication by Oxford University Press and a DOI has been pre-registered. A preprint version is available at: <https://arxiv.org/abs/2504.11340g>

S. Ramakrishnan, A.V. Patnaik, G. Grillo, L. Tubiana, D. Michieletto Oxford University Press (OUP) S. Ramakrishnan, A.V. Patnaik, G. Grillo, L. Tubiana, D. Michieletto, arXiv:2504.11340 (2025). Nucleic Acids Research (NAR)